INCH-POUND
A-A-50431B
October 24, 2016
SUPERSEDING
A-A-50431A
November 7, 1991

COMMERCIAL ITEM DESCRIPTION

COUPLINGS – QUICK DISCONNECT, BOWES TYPES (AIR HOSES AND PIPING)

The General Services Administration has authorized the use of this commercial item description for all federal agencies.

1. SCOPE. This commercial item description (CID) covers quick-acting, self-locking couplings for Bowes types pneumatic hose and piping connections where rapid coupling and uncoupling are required. Bowes types quick disconnect couplings are not suitable for use in fuel services or where fluids have a specific gravity less than 1.00. Typical services for Bowes types quick disconnect couplings are air, inert gases, and water.

2. CLASSIFICATION. Coupling halves and fittings shall be of the following classes and sizes:

2.1 Class.

2.1.1 Class H.M. Male coupling half with hose end for ⅜-inch, 7/16-inch, ½-inch, ⅝-inch, ¾-inch, 1-inch, and 1¼-inch hose.

2.1.2 Class H.F. Female coupling half with hose end for ⅜-inch, 7/16-inch, ½-inch, ⅝-inch, ¾-inch, 1-inch, and 1¼-inch hose.

2.1.3 Class O.M. Male coupling half with pipe end (external thread) for ⅜-inch, 7/16-inch ½-inch, ¾-inch, 1-inch, or 1¼-inch pipe.

2.1.4 Class O.F. Female coupling half with pipe end (external thread) for ⅜-inch, 7/16-inch ½-inch, ¾-inch, 1-inch, or 1¼-inch pipe.

2.1.5 Class I.M. Male coupling half with pipe end (internal thread) for ⅜-inch, 7/16-inch, ½-inch, ¾-inch, 1-inch, or 1¼-inch pipe.

2.1.6 Class I.F. Female coupling half with pipe end (internal thread) for ⅜-inch, 7/16-inch, ½-inch, ¾-inch, 1-inch, or 1¼-inch pipe.

3. SALIENT CHARACTERISTICS.

3.1 Materials. Fittings shall be made of material in accordance with ASTM B61 bronze (UNS no. C92200), ASTM A582/A582M corrosion resistant steel (UNS no. S30300), or ASTM A108 steel (UNS no.G12144 or G12150). Locking springs, retaining clips, and retaining rings shall collectively be made of phosphor-bronze or corrosion resistant steel (UNS no. S30200). Gaskets shall be made of nitrile or alternative materials that will ensure system compatibility.

Beneficial comments, recommendations, additions, deletions, clarifications, etc. and any data that may improve this document should be sent to: Commander, Naval Sea Systems Command, ATTN: SEA 05S, 1333 Isaac Hull Avenue, SE, Stop 5160, Washington Navy Yard DC 20376-5160 or emailed to CommandStandards@navy.mil, with the subject line "Document Comment". Since contact information can change, you may want to verify the currency of this address information using the ASSIST Online database at https://assist.dla.mil.

FSC 4730

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3.2 Interchangeability. Coupling ends of all coupling halves and fittings for use with ⅜-inch, 7/16-inch, ½-inch, ⅝-inch, ¾-inch, and 1-inch hose, as well as ½-inch or ¾-inch pipe, shall be completely interchangeable, lock effectively, and maintain an airtight seal.

Coupling ends of all couplings for use with 1-inch or 1¼-inch hose and 1-inch or 1¼-inch pipe shall be completely interchangeable, lock effectively, and maintain an airtight seal.

Where tolerances are not specifically mentioned, a tolerance of ±0.005 inch will be permitted in all dimensions for which such variations will not affect interchangeability.

3.3 Coupling assembly and details. Coupling assembly and details shall conform to figure 1.

3.4 Description of classes.

3.4.1 Class H.M. The coupling half shall have a male Bowes type coupling half (fitted with self-locking clutch including spring, spring clip, and clutch-retaining ring) at one end, and a serrated hose shank at the other end, all conforming to figures 1, 2, and 3.

3.4.2 Class H.F. The coupling half shall have a female Bowes type coupling half (fitted with a U-shaped rubber gasket with lip facing hose shank) at one end, and a serrated hose shank at the other end, all conforming to figures 1, 2, and 3.

3.4.3 Class O.M. The coupling half shall have a male Bowes type coupling half (fitted with self-locking clutch including spring, spring clip, and clutch-retaining ring) at one end, and the opposite end shall be threaded with American National Standard taper pipe threads, all conforming to figures 1, 2, and 3.

3.4.4 Class O.F. The coupling half shall have a female Bowes type coupling half (fitted with a U-shaped rubber gasket with lip facing hose shank) at one end, and the opposite end shall be threaded with American National Standard taper pipe threads, all conforming to figures 1, 2, and 3.

3.4.5 Class I.M. The coupling half shall have a male Bowes type coupling half (fitted with self-locking clutch including spring, spring clip, and clutch-retaining ring) at one end and the opposite end shall be threaded with American National Standard taper pipe threads, all conforming to figures 1, 2, and 3.

3.4.6 Class I.F. The coupling half shall have a female Bowes type coupling half (fitted with a U-shaped rubber gasket with lip facing hose shank) at one end, and the opposite end shall be threaded with American National Standard taper pipe threads, all conforming to figures 1, 2, and 3.

3.5 Marking. Each coupling half or fitting shall be marked in a plain and permanent manner with the manufacturer's name or with a trademark of such known character that the source of manufacture can be readily determined.

3.6 Workmanship. Exposed surfaces shall be free from projections that may cause the coupling to unlock when the hose is dragged over uneven surfaces. Couplings shall be produced to be free from rough surfaces that might injure the hands of the operator, except that the locking clutch shall be knurled and grooved or otherwise constructed to ensure a firm grip for the fingers.

3.7 Performance. The coupling halves and fittings shall have the following performance characteristics:

a. Couplings shall withstand, without leakage or any other indication of failure, a hydrostatic pressure test of 1½ times the working pressure for 5 minutes.

b. The compressive force to operate the couplings shall be 15 to 19 pounds. Samples shall be measured for compliance with this requirement.

3.8 Steel fitting finish (except corrosion resistant steel). All plating shall be capable of meeting a 500-hour minimum salt spray test in accordance with ASTM B117. The fittings shall show no evidence of corrosion after 500 hours of salt spray.

4. REGULATORY REQUIREMENTS. The offeror/contractor is encouraged to use recovered materials to the maximum extent practicable, in accordance with paragraph 23.403 of the Federal Acquisition Regulation (FAR).

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5. PRODUCT CONFORMANCE PROVISIONS.

5.1 Product conformance. The products provided shall meet the salient characteristics of this Commercial Item Description, conform to the producer's own drawings, specifications, standards, and quality assurance practices, and be the same product offered for sale in the commercial marketplace. The government reserves the right to require proof of such conformance.

5.2 Lot. For the purpose of inspection and testing, a lot shall consist of the same class and size couplings from the same lot of raw materials, produced under essentially the same conditions, and offered for delivery at one time.

Sample fittings shall be selected from each lot in accordance with ANSI/ASQ Z1.4 at inspection level II. The samples selected shall be subjected to the performance requirements of 3.7 and shall be visually and dimensionally examined to verify compliance with this CID. If any sample fails to conform, the lot shall be subject to rejection.

5.3 Market acceptability. The following market acceptability criteria are necessary to document the quality of the product to be provided under this CID:

a. The item offered must have been sold to the government or commercial market.

b. A commercial item is required to ensure serviceability, reliability, and quality of material.

c. A market tested item is required to preclude untested or experimental units.

6. PACKAGING. Preservation, packing, and marking shall be as specified in the contract or order.

7. NOTES.

7.1 Part or identification number (PIN). The following PIN procedure is for government purposes and does not constitute a requirement for the contractor. The PIN to be used for quick disconnect couplings acquired to this CID is as follows (see 7.3):

AA50431-XXXXXXX

- Gasket Material
  - 0 – Nitrile
  - 1 – Other
- Connection
  - 0 – Hose
  - 1 – Pipe
- Size
  - A – ⅜ inch
  - B – 7/16 inch
  - C – ½ inch
  - D – ⅝ inch
  - E – ¾ inch
  - F – 1 inch
  - G – 1¼ inches
- Fitting Material
  - BZ – Bronze
  - CR – Corrosion Resistant Steel
  - ST – Steel
- Class
  - HM – H.M.
  - HF – H.F.
  - OM – O.M.
  - OF – O.F.
  - IM – I.M.
  - IF – I.F.
- CID Number (without revision)

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7.1.1 Example of PIN. The PIN AA50431-HMBZA00 specifies an H.M. (male coupling half), bronze, ⅜-inch, hose-connected fitting with a nitrile gasket.

7.2 Source of documents.

7.2.1 American Society for Quality (ASQ). ASQ standards are available from the American Society for Quality, 600 North Plankinton Ave., Milwaukee, WI 53203 or online at www.asq.org.

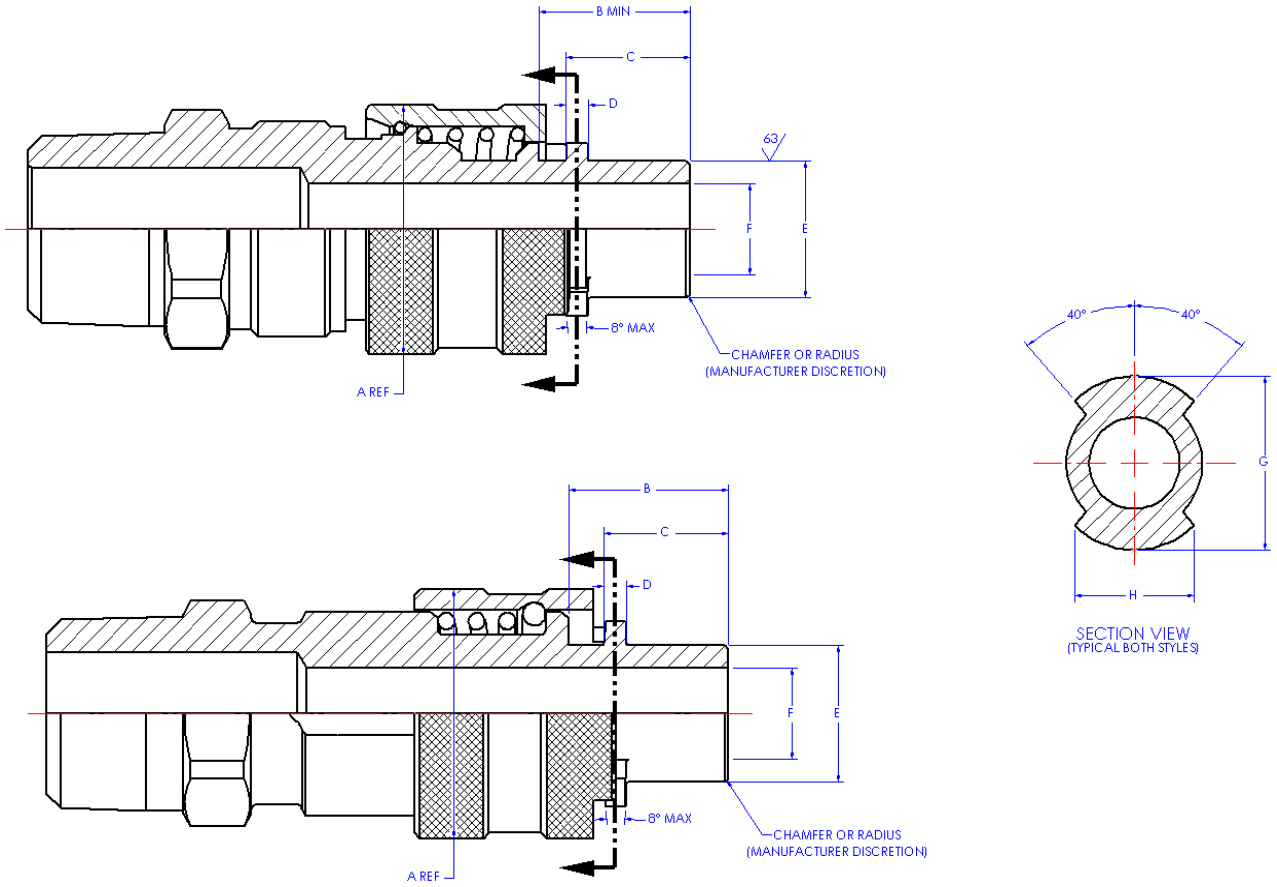
7.2.2 ASTM International. ASTM standards are available from ASTM International, 100 Barr Harbor Dr., P.O. Box C700, West Conshohocken, PA 19428-2959 or online at www.astm.org.

7.2.3 FAR. The Federal Acquisition Regulation may be obtained from the Superintendent of Documents, U.S. Government Printing Office, Washington, DC 20402 or online at https://www.acquisition.gov/far.

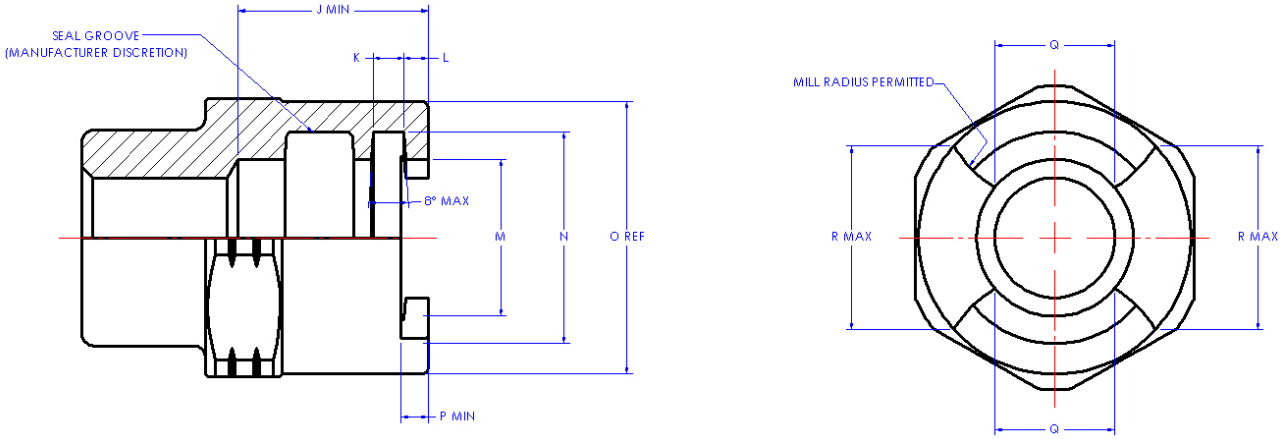
7.3 Ordering data. The contract or order should specify the following:

a. CID document number, revision, and CID PIN (see 7.1).
b. Quantity, class, and size required (refer to figure 2 for the appropriate nominal pipe or hose size).
c. Product conformance provisions.
d. Packaging requirements.

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Plug Dimensions



Socket Dimensions

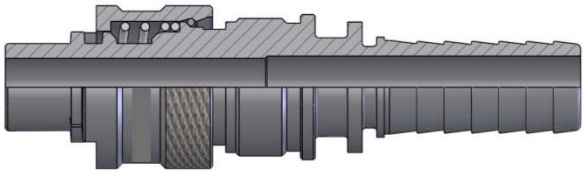
FIGURE 1. Coupling details.

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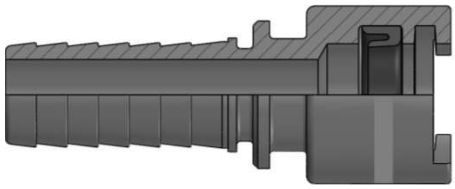
Dimension	Nominal Size of Hose or Pipe (inches)	
	⅜, ½, ¾, 1	1¼
A	1.375	1.875
B	0.830	1.250
C	0.685	0.970
D	0.125	0.188
E	0.750	1.281
F	0.500	1.000
G	0.985±0.040	1.575±0.010
H	0.690±0.040	1.063±0.010
J	0.930	1.350
K	0.150	0.188
L	0.120	0.156
M	0.775	1.313
N	1.045	1.625
O	1.375	1.875
P	0.125	0.160
Q	0.590±0.035	1.020±0.010
R	0.930	1.320

FIGURE 2. List of dimensions.

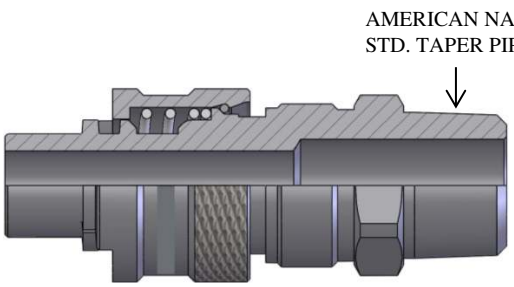
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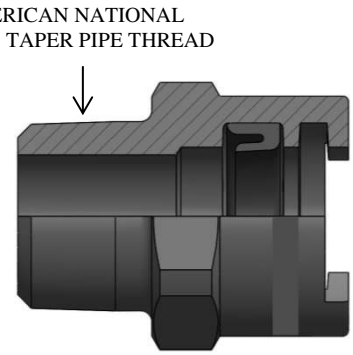
CLASS H.M. MALE COUPLING HALF WITH HOSE END



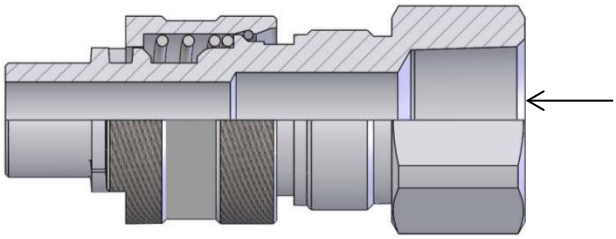
CLASS H.F. FEMALE COUPLING HALF WITH HOSE END



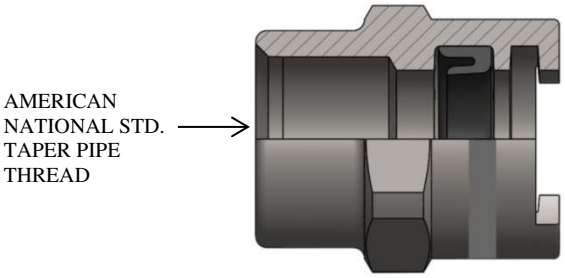
CLASS O.M. MALE COUPLING HALF WITH PIPE END



CLASS O.F. FEMALE COUPLING HALF WITH PIPE END



CLASS I.M. MALE COUPLING HALF WITH PIPE END



CLASS I.F. FEMALE COUPLING HALF WITH PIPE END

FIGURE 3. Classes.

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MILITARY INTERESTS:

Custodians:
Army – AR
Navy – SH
DLA – CC

Review Activity:
Navy – YD

CIVIL AGENCY COORDINATING ACTIVITY:
GSA – FAS

Preparing Activity:
Navy – SH
(Project 4730-2016-043)

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